Multi-Objective Spatial Optimization Results

Balancing global dispersion, local clustering, and algorithmic fairness on combinatorial spatial domains

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1 Section 1: The Spatial Pathology (The "Why")

Global fairness metrics such as Jain's Fairness Index (JFI) are blind to local structural defects. We demonstrate that JFI and Moran's I are orthogonal objectives on the hexagonal lattice.

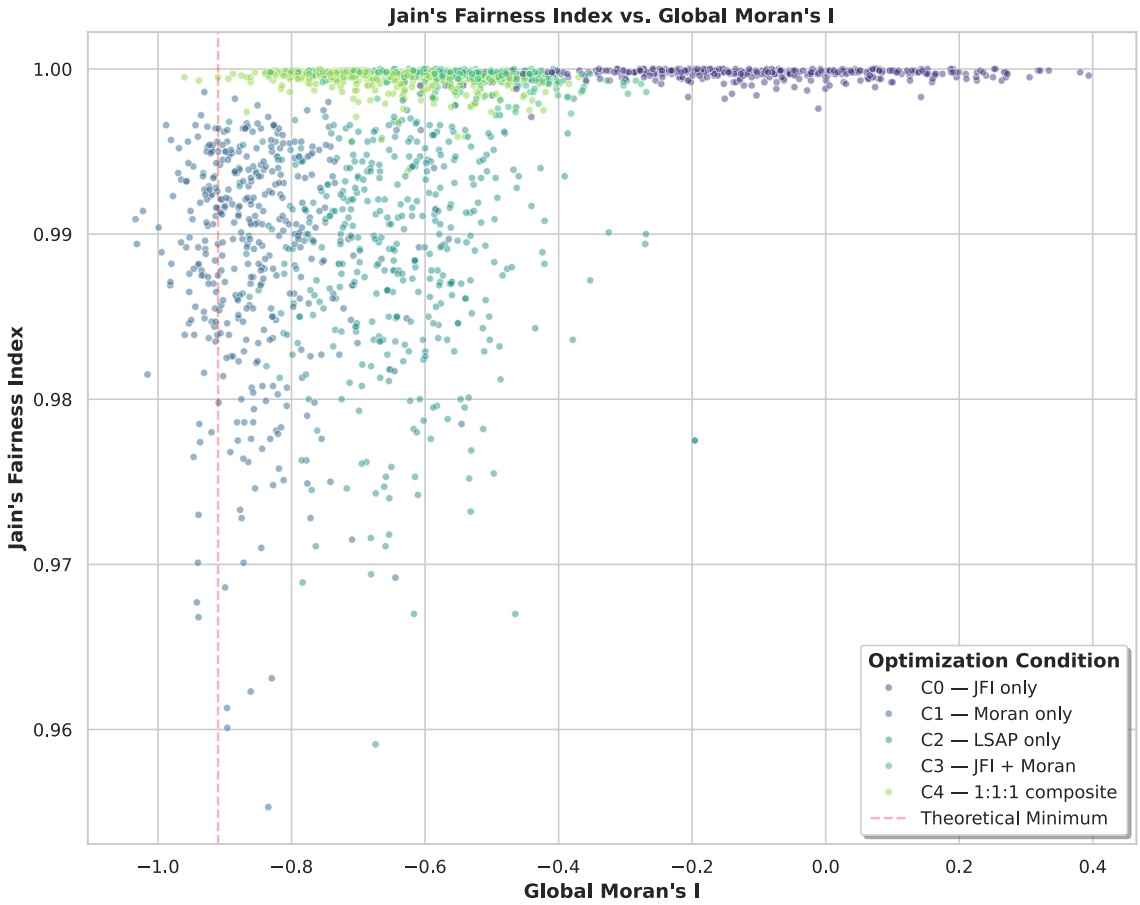

Figure 1: Tradeoff between JFI and Global Moran's I (Phase 1). The scatter plot reveals a wide vertical spread: an optimizer can drive Moran's I to its theoretical minimum (~ −0.91) while maintaining near-perfect fairness (JFI > 0.999). Global metrics average away local clustering.

1.1 Pathological Fairness: Visualizing Residuals

To prove the failure of global metrics, we examine a "matched pair": two maps with identical JFI but vastly different local structures. We visualize the **spatial delta** between a fairness-only baseline and a Moran-optimized configuration.

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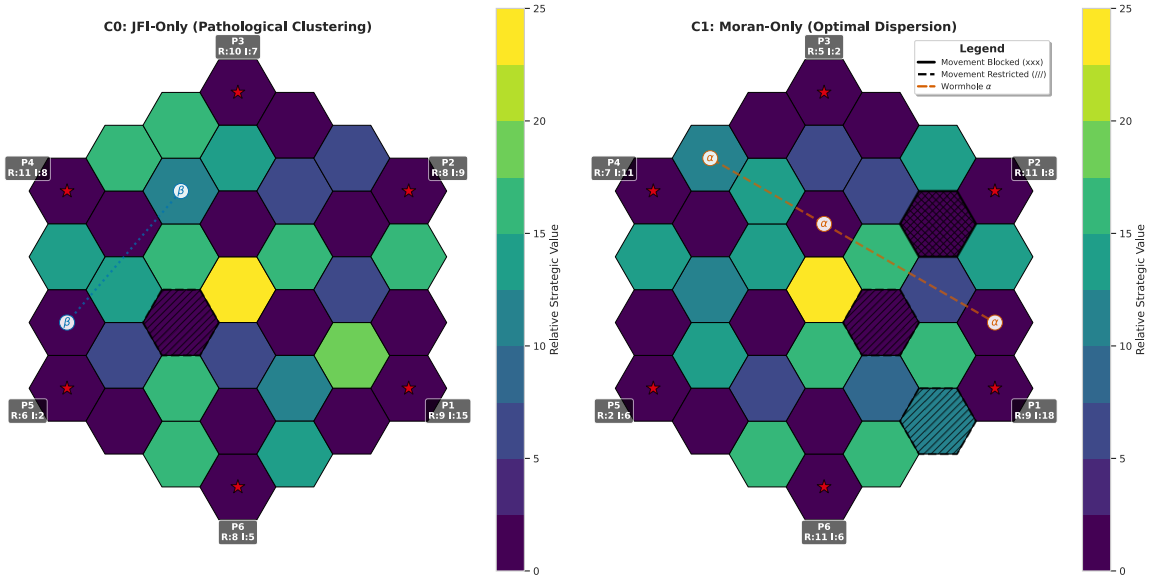
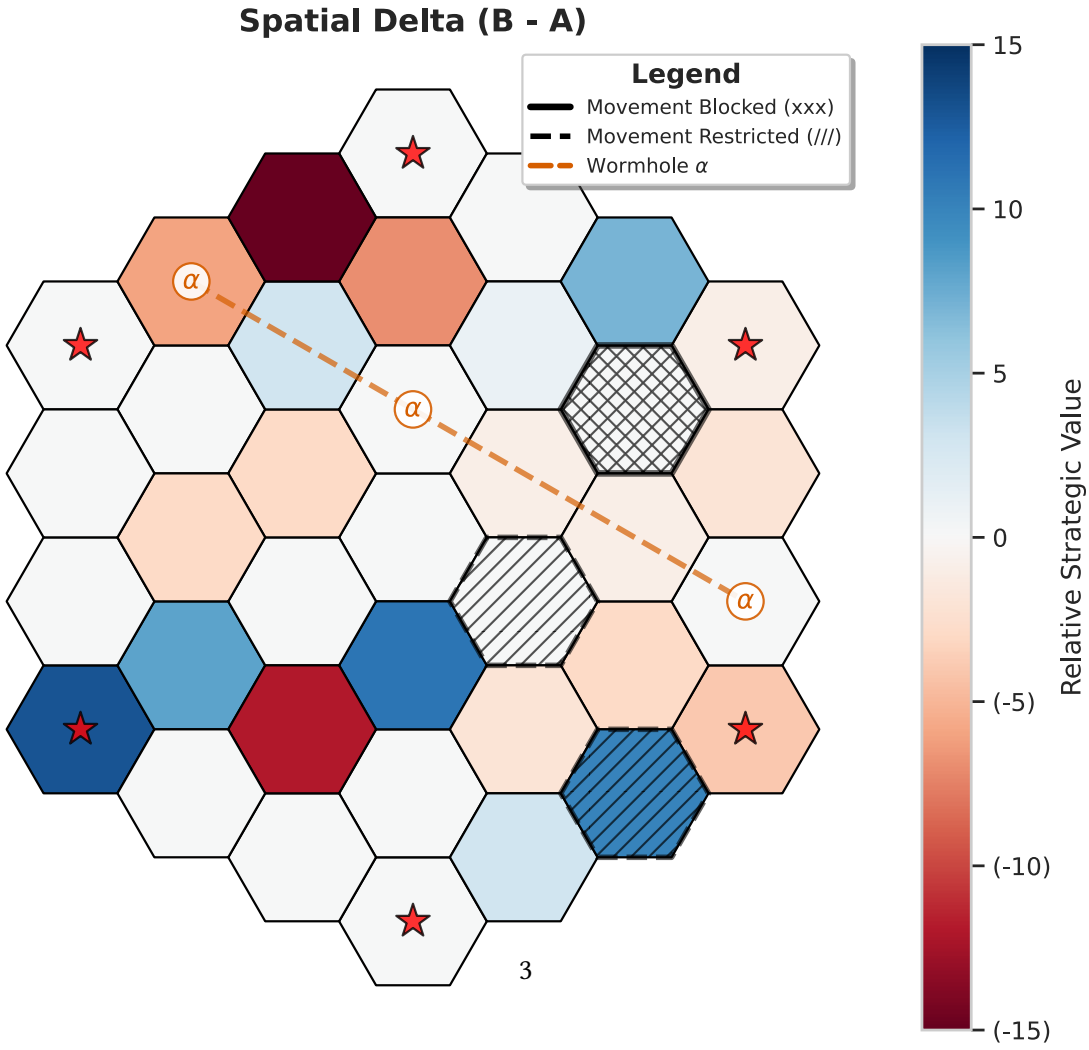


(a) Spatial Residuals: Comparing C0 (JFI-only) vs C1 (Moran-only). The delta map highlights the localized redistribution of value. Blue hexes indicate gained strategic value, while red hexes indicate broken clusters.



(b)

1.2 Uncertainty Quantification: Induced Pathologies

To understand how optimizing *only* for global fairness actively induces pathological local structures, we compare the Local Spatial Association Penalty (LSAP) of purely random maps versus JFI-only optimized maps.

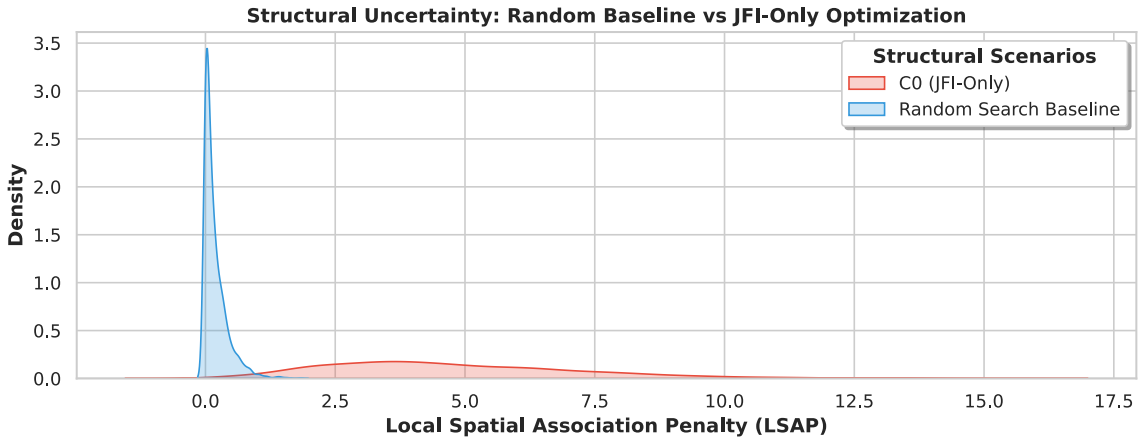


Figure 3: Distribution of LSAP scores. Random maps exhibit near-zero structural penalties, while JFI-only optimization drastically inflates local structural defects (uncertainty).

2 Section 2: The Objective Space & Convergence (The "How")

2.1 The Objective Function

The optimizer minimizes a scalar composite equation balancing three competing spatial priorities:

$$S = \frac{1}{3}h(I) + \frac{1}{3}(1 - J_{min}) + \frac{1}{3}\frac{LSAP}{n(n-1)}$$

where $h(I) = max(0, I - E[I])$.

The multi-objective problem defines a Pareto frontier where global dispersion, local clustering, and fairness are in competition.

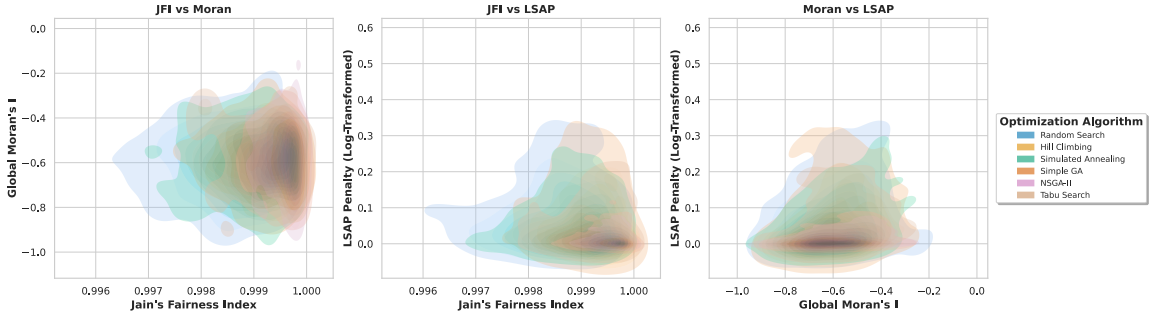


Figure 4: Pareto Projections (Algorithm Comparison). 2D KDE contour plots reveal where the mass of the algorithmic convergence lies, moving away from overplotted scatters. The Utopia Point indicates the theoretical unachievable optimum.

2.2 3D Pareto Surface

To reveal geometric features hidden by 2D projections, we visualize the full 3D objective space.

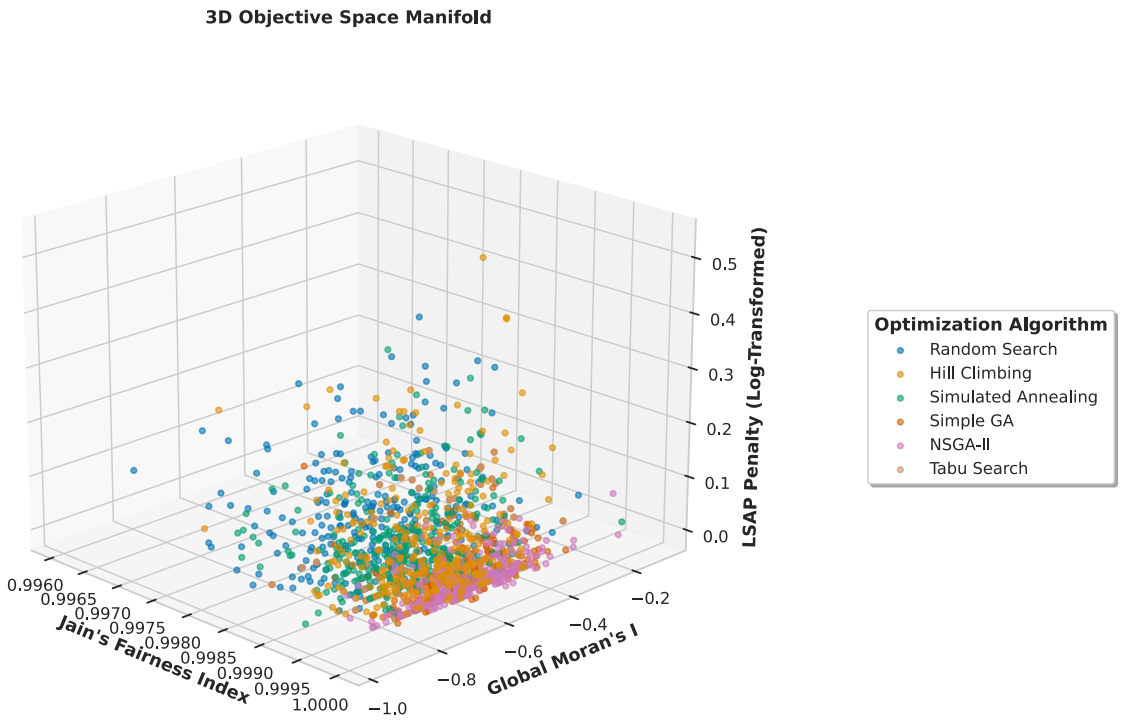


Figure 5: 3D Pareto Surface. Rotating the objective space reveals the true shape of the combinatorial manifold and the strict trade-offs between JFI, Moran's I, and LSAP.

2.3 Computational Efficiency

Convergence trajectories prove that Simulated Annealing (SA) provides the most efficient path through the combinatorial manifold.

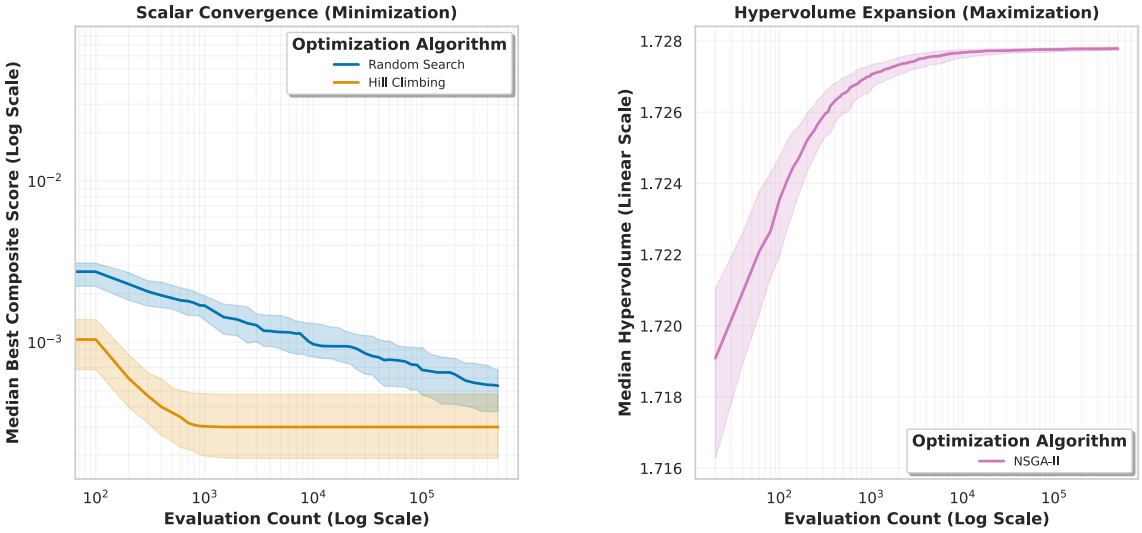


Figure 6: Algorithmic Efficiency and Frontier Expansion. (Left) Median Convergence Trajectories for single-objective algorithms tracking the scalar composite score. (Right) Hypervolume Convergence for NSGA-II tracking the expanding Lebesgue measure of the dominated objective space. By placing these mathematically distinct domains on a shared logarithmic evaluation axis, we isolate their performance envelopes without conflating their codomains. Shaded regions represent the Interquartile Range (IQR) across all seeds.

3 Section 3: The Optimized Spatial Manifold (The "Result")

The final composite objective (1:1:1 JFI:Moran:LSAP) successfully cures the spatial pathology with negligible cost to fairness.

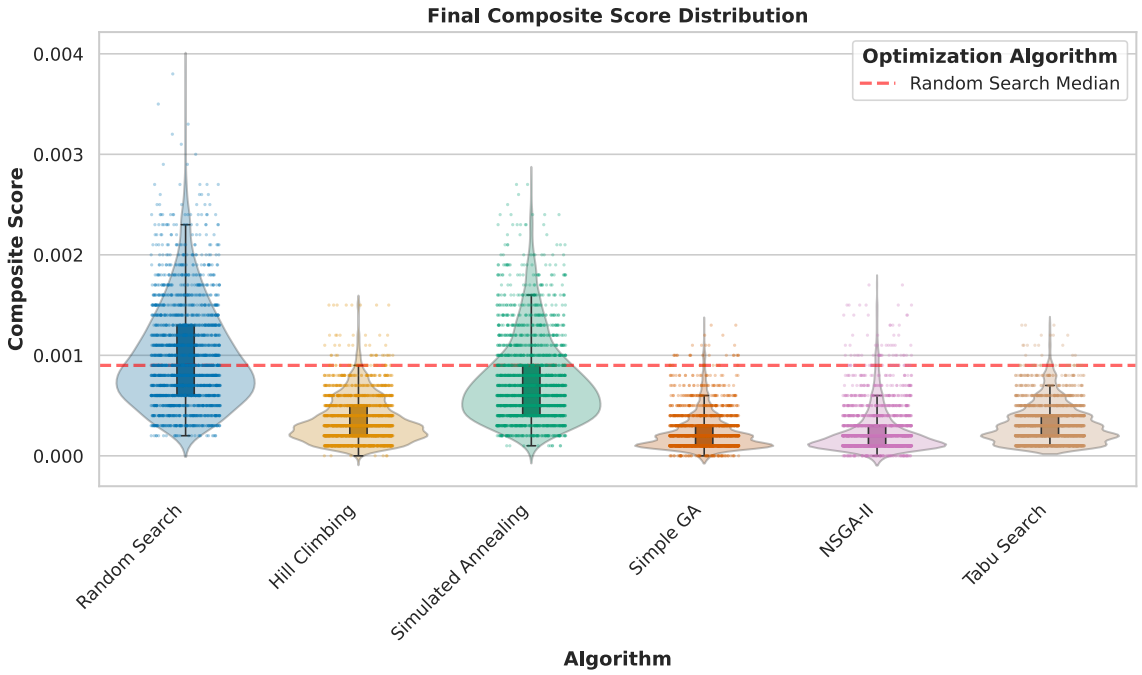
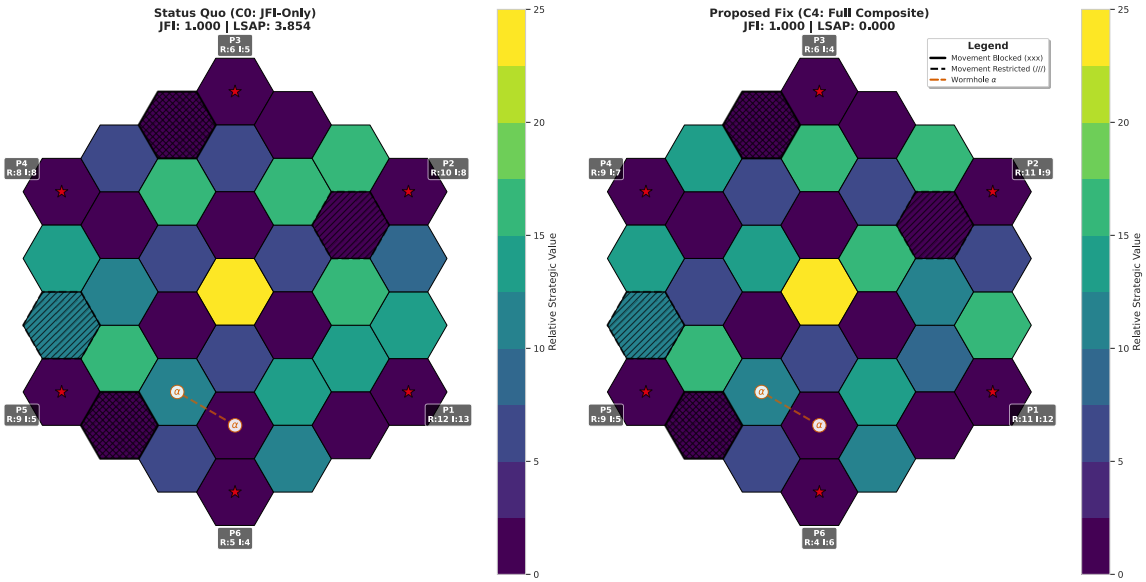


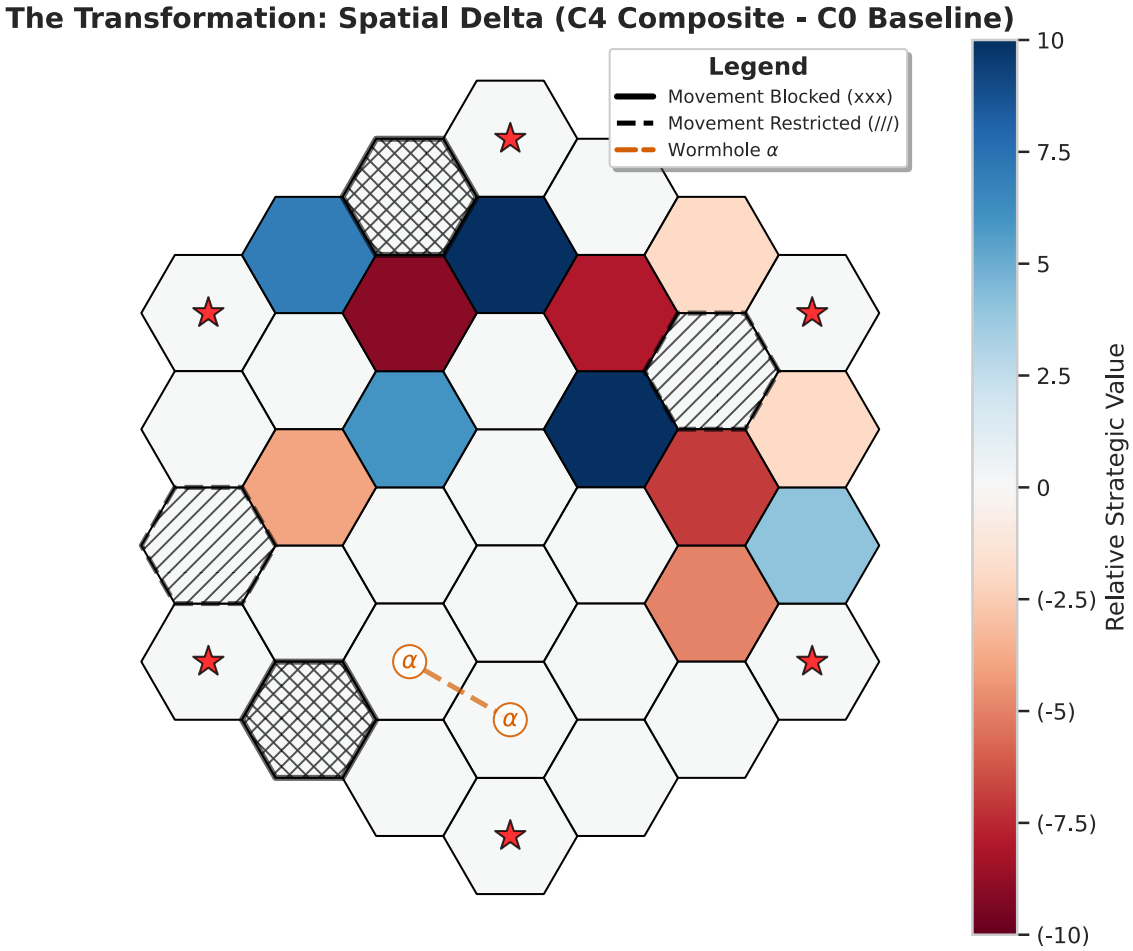
Figure 7: Final Composite Score Distribution. The raincloud plot combines the probability density, boxplot medians, and raw jittered data to prove statistical robustness. The dashed red line represents the Random Search baseline median.

3.1 The Optimal State

The following configuration represents the absolute best-performing map found by the Simulated Annealing solver on the composite landscape, compared against a JFI-only baseline generated from the same starting seed.



(a) The Proposed Solution: Matched-seed comparison between the Status Quo (JFI-only) and the Full Composite objective. By holding the seed and the solver constant, the spatial delta isolates the exact structural improvements driven by the LSAP and Moran penalties.



(b)

Figure 8:

Effect Sizes (Vargha-Delaney A test): Non-parametric statistical tests reveal a 0.904 probability that the C3 configuration (JFI + Moran) contains more local clustering anomalies than the fully optimized C4 composite map.

4 Section 4: Discussion & Future Work

4.1 Geospatial AI Context

A Twilight Imperium map functions as a discrete spatial domain. When we measure Moran's I and LSAP, we are fundamentally executing edge-detection and local texture analysis, conceptually identical to analyzing pixel neighborhoods in satellite imagery to detect harmful algal blooms or coastal boundaries. This framework effectively isolates the local spatial pathologies that global aggregation masks.

4.2 Multi-Objective Tension

The optimization Pareto frontier established here is rooted in Real Analysis. Navigating the boundary of a feasible set in this high-dimensional combinatorial manifold is identical to the multi-objective tension encountered in optimizing Neural Radiance Fields (NeRFs) or other hardware-constrained models: minimizing computational cost (latency) while maximizing structural fidelity (reconstruction accuracy). Future work will explore adapting this framework to Graph Neural Networks (GNNs) that natively respect adjacency constraints without requiring scalarized penalty terms.